Computing the Nearest Correlation Matrix: A Problem from Finance

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Our Goal

Given a symmetric matrix $A \in \mathbb{R}^{n \times n}$, we are interested in computing the nearest correlation matrix to A, i.e. we are looking for a matrix X which

- is positive semidefinite;
- is symmetric;
- has unit diagonal;

that verifies

$$\boxed{\min \|A - X\|}$$

for some specific norm.

Why such a big deal?

This problem arises when the data from which the correlations are constructed is asynchronous or incomplete, or when models are stress-tested by artificially adjusting individual correlations.

About Norms

We will use the weighted Frobenius norm, i.e.

$$\|A\|_W = \|W^{\frac{1}{2}} A W^{\frac{1}{2}}\|_F$$

where W is a symmetric positive definite matrix.

Convex Sets

We recall the following

Definition (Convex sets)

A subset C of $\mathbb{R}^m$ is said to be convex if for every x, y in C and for every t in $[0, 1]$

$$tx + (1 - t)y \in C.$$

The Dykstra's algorithm

Let H be a Hilbert space, $g \in H$, $K_1, K_2 \subset H$ closed and convex

Algorithm 1 Dykstra's algorithm

$g_0 = g, \quad I_1 = I_2 = 0$
for $i = 1, 2, \ldots,$ maxit **do**
$g_i = P_{K_{i \bmod 2}}(g_{i-1} - I_{i \bmod 2})$
$I_{i \bmod 2} = g_i - (g_{i-1} - I_{i \bmod 2})$
end for

Dykstra's Algorithm Convergence

Theorem

The sequence $\{g_n\}_{n=0}^{\infty}$ *converges to* $g^* \in H$, *where*

$$g^* = \arg \min_{x \in K_0 \cap K_1} \|g - x\|.$$

Dykstra's Algorithm Convergence

Theorem (Deutsch and Hundal)

The rate of convergence of Dykstra's algorithm for polyhedral set is linear.

In the nonpolyhedral case, as far as we know, no general convergence results has been established.

Dykstra's Hypotesis

In order to follow Dykstra's algorithm we must check if we are in the right setting.

1. Does $\|\cdot\|_W$ come from an inner product?
2. Is $(\mathbb{R}^{n\times n}, \|\cdot\|_W)$ complete?
3. Is the set of correlation matrices a finite intersection of closed convex sets?

Is $S \cap U$ a convex set?

We define

$$S = \{Y \in \mathbb{R}^{n \times n} : Y \text{ is symmetric and positive semidefinite}\}$$

$$U = \{Y \in \mathbb{R}^{n \times n} : Y \text{ is symmetric and has unit diagonal}\}$$

The set of correlation matrices is given by $S \cap U$.

Projecting onto convex sets

Dykstra's algorithm requires projecting onto convex sets. We look for the projections onto S and U.

Theory

For norms induced by inner products we can use the following

Theorem

Let x be a vector in a Hilbert space H and let K be a closed convex subset of H. Then there is an unique vector k_0 in K such that

$$\|x - k_0\| \leq \|x - k\|$$

for all k in K. Furthermore, a necessary and sufficient condition that k_0 be the unique minimizing vector is that $\langle x - k_0, k - k_0 \rangle \leq 0$ for all k in K.

Projecting onto U

Theorem

For the W - norm,

$$P_U(A) = A - W^{-1}\mathrm{diag}(\theta_i)W^{-1},$$

where $\theta = [\theta_1, \cdots, \theta_n]^T$ is the solution of the linear system

$$(W^{-1} \circ W^{-1})\theta = \mathrm{diag}(A - I).$$

Projecting onto S

For a symmetric $A \in \mathbb{R}^{n \times n}$ with spectral decomposition $A = QDQ^T$, where $D = \mathrm{diag}(\lambda_i)$ and Q is orthogonal, let

$$A_+ = Q\mathrm{diag}(\max(\lambda_i, 0))Q^T, \quad A_- = Q\mathrm{diag}(\min(\lambda_i, 0))Q^T.$$

We have $A = A_+ + A_-, A_+A_- = A_-A_+ = 0$.

Projecting onto S

Theorem

For the W - norm,

$$P_S(A) = W^{-\frac{1}{2}}((W^{\frac{1}{2}}AW^{\frac{1}{2}})_+)W^{-\frac{1}{2}}.$$

The projection onto U

Algorithm 2 Projection $P_U(A)$ for the W-norm (SPD W)

Require: $A \in \mathbb{R}^{n\times n}$, $W \in \mathbb{R}^{n\times n}$ symmetric positive definite

Ensure: $P_U(A)$

1: Compute the Cholesky factorization $W = R^T R$
2: Compute $M = W^{-1}$ by triangular solves: $M = R^{-1}R^{-T}$
3: $r = \mathrm{diag}(A - I)$
4: Form $H = M \circ M$
5: Solve the linear system $H\,\theta = r$
6: Form $Z = M\,\mathrm{diag}(\theta)$ ▷ scale columns: $Z_{:,j} = \theta_j M_{:,j}$
7: $P_U(A) = A - Z\,M$

MATLAB implementation of $P_U(A)$

```
function PU = Pu(A, W)
    n = size(A,1);
    R = chol(W);
    M = R \ (R' \ eye(n));
    rhs = diag(A) - 1;
    theta = (M.^2) \ rhs;
    PU = A - ( (M .* theta.') * M );
end
```

The projection onto S

Algorithm 3 Projection $P_S(A)$ for the W-norm

Require: $A \in \mathbb{R}^{n \times n}$ symmetric, $W \in \mathbb{R}^{n \times n}$ symmetric positive definite

Ensure: $P_S(A)$

1: Compute the Cholesky factorization $W = R^T R$
2: Form $B = RAR^T$
3: Symmetrize $B = (B + B^T)/2$
4: Compute the spectral decomposition $B = Q\Lambda Q^T$
5: Set $\Lambda_+ = \max(\Lambda, 0)$
6: Form $B_+ = Q\Lambda_+ Q^T$
7: $P_S(A) = R^{-1} B_+ R^{-T}$

MATLAB implementation of $P_S(A)$

```
function PS = Ps(A, W)
    R = chol(W);
    B = R * A * R';
    B = (B + B')/2;
    [Q, D] = eig(B);
    Dp = max(D, 0);
    Bp = Q * Dp * Q';
    PS = R \ (Bp / R');
    PS = (PS + PS')/2;
end
```

Higham's implementation

Algorithm 4 Higham's implementation (Dykstra's algorithm)

$\Delta S_0 = 0, \quad Y_0 = A$

for $k = 1, 2, \ldots$ **do**

$R_k = Y_{k-1} - \Delta S_{k-1}$

$X_k = P_S(R_k)$

$\Delta S_k = X_k - R_k$

$Y_k = P_U(X_k)$

end for

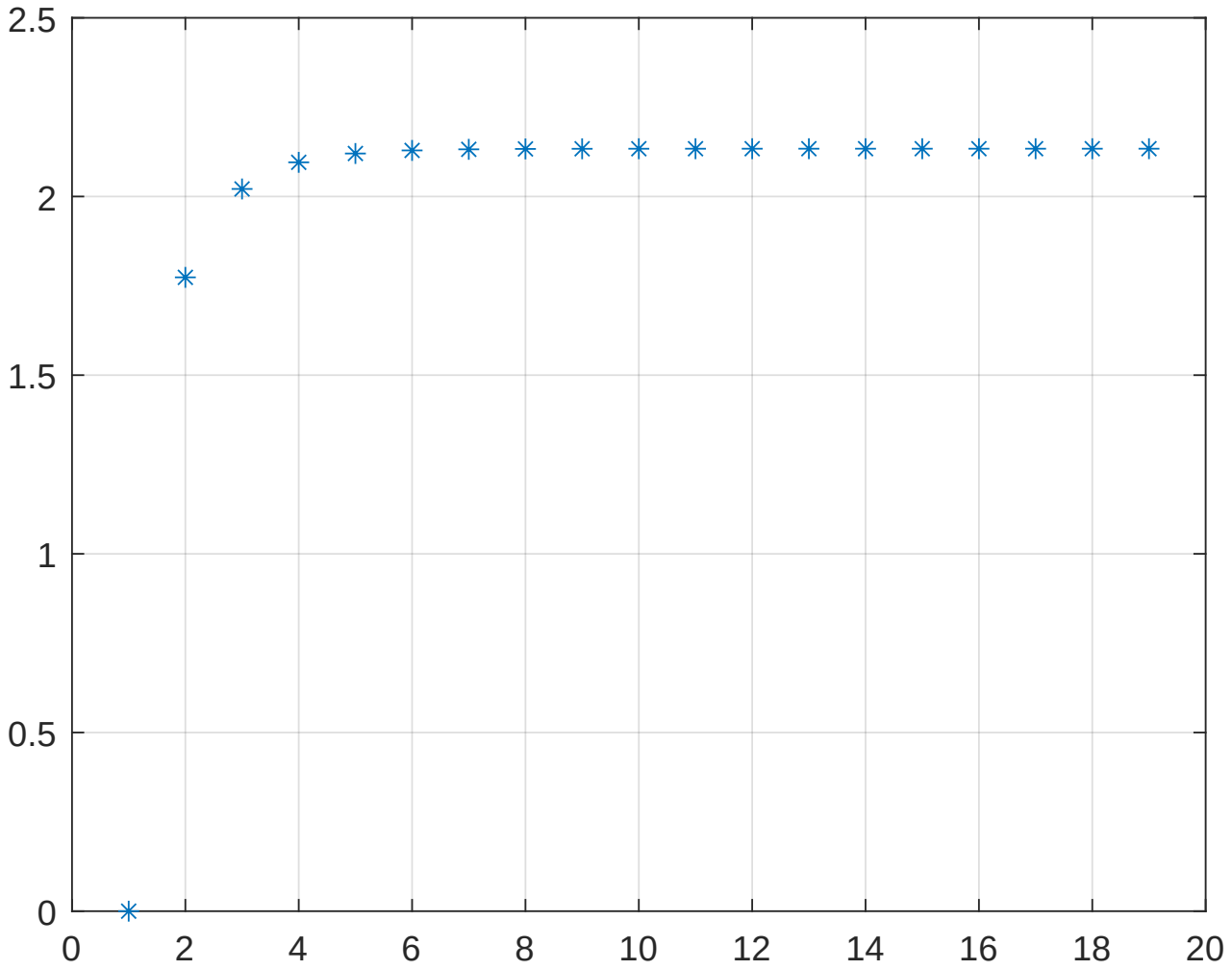
Experiment 1

In our first example we take the positive definite matrix

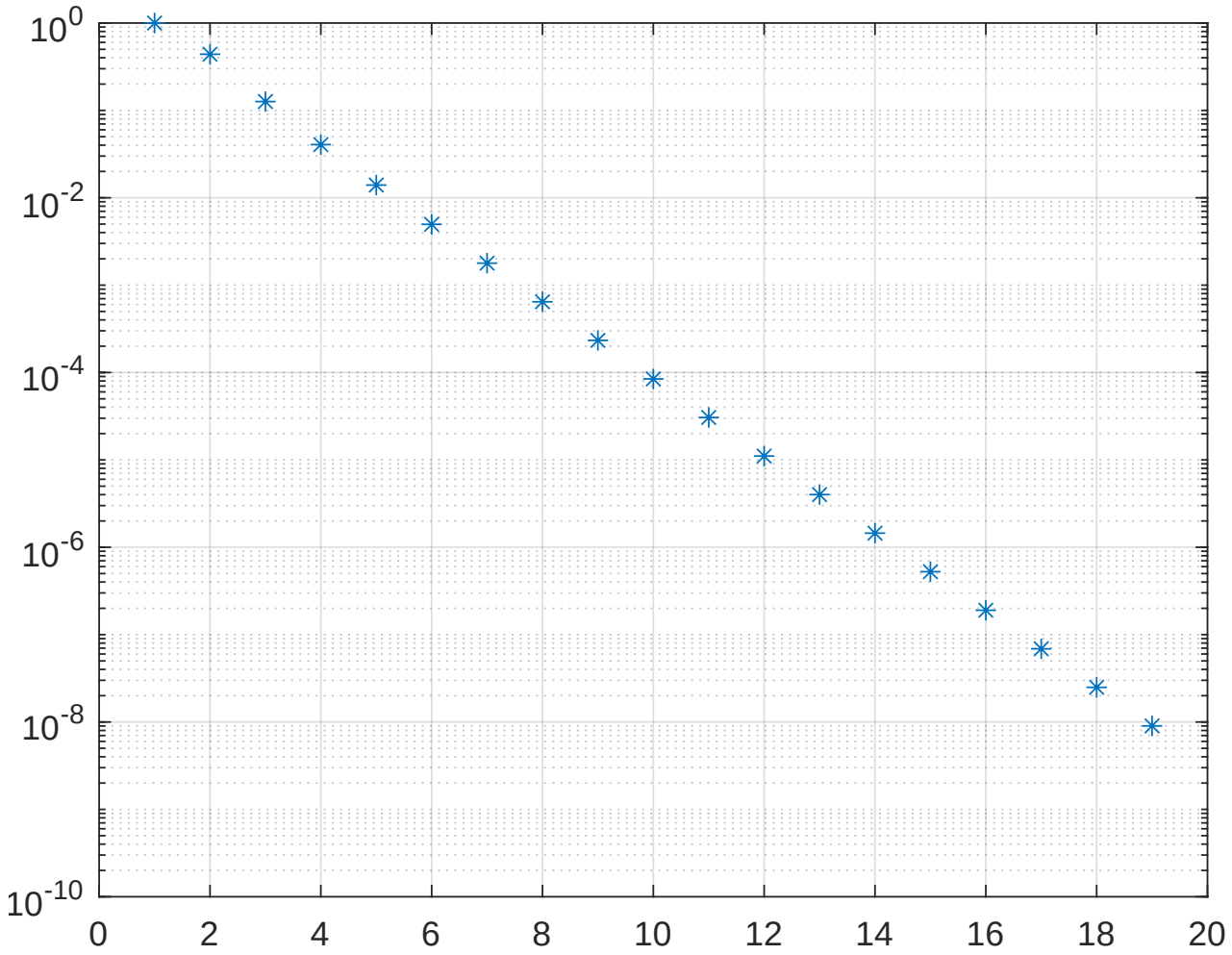
$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

with $\mathrm{tol} = 10^{-8}$, $W = \mathrm{Id}$.

$\|A - X_k\|_W$ Residue



$\|X_k - X_{k-1}\|_W$ Residue



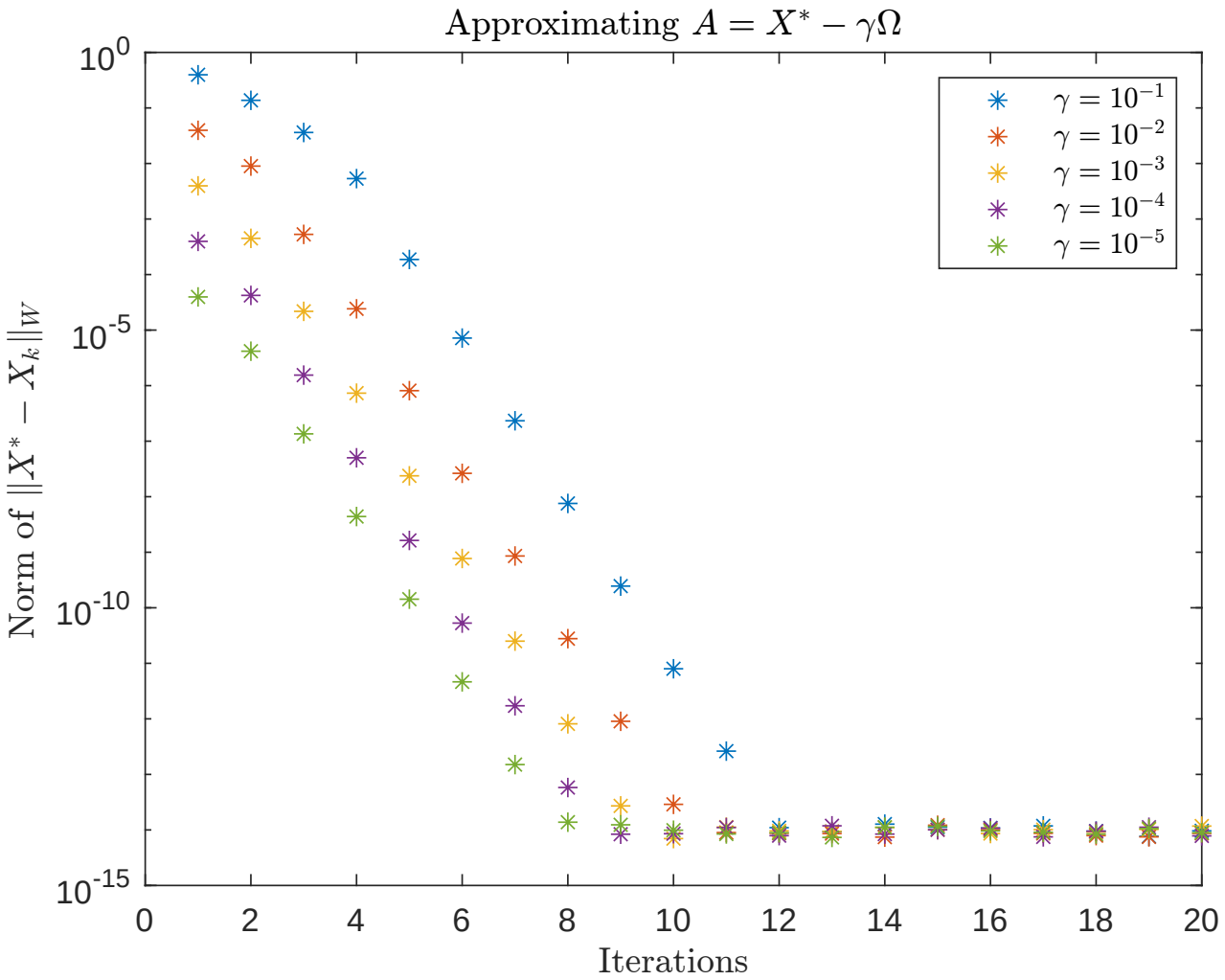
Experiment 2

We sample a random correlation matrix X^* and introduce the following perturbation:

$$A = X^* - \gamma \lambda_{\max}(X^*) I, \qquad \gamma = 10^i, \qquad i = 1, \ldots, 5.$$

Then we compute the nearest correlation matrix to A.

Experiment 3



Thank you for your attention!